

C655

Zoology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

At a public aquarium that opened in 1998, a volunteer shows visitors a jellyfish tank next to a flatworm display. The jellyfish tank holds 900 liters of seawater. Which statement correctly compares the two body plans?

- A. The jellyfish is radially symmetric; the flatworm is bilaterally symmetric with a head end.
- B. The jellyfish is bilateral and the flatworm is radial.
- C. Neither has symmetry, since both live in water.
- D. Both are bilaterally symmetric because both are invertebrates.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A teacher sets out three preserved specimens bought from a supply company in the spring: a planarian (flatworm), a roundworm, and an earthworm. Which list correctly names each one's body cavity type?

- A. All three are coelomates because they are all worms.
- B. Acoelomate, pseudocoelomate, coelomate
- C. Coelomate, pseudocoelomate, acoelomate
- D. Acoelomate, coelomate, coelomate

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

An animal has a fluid-filled body cavity lined only partly by mesoderm. Which animal fits?

- A. A sponge
- B. A planarian (flatworm)
- C. An earthworm
- D. A nematode (roundworm)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A student notices that sea stars are invertebrates with a spiny skin, and asks which group they are most closely related to. The class recently watched a documentary about coral reefs. Based on embryo development, what is the best answer?

- A. Corals, because both live on reefs.
- B. Mollusks, because in both the blastopore becomes the mouth.
- C. Arthropods, because both are invertebrates with hard outer parts.
- D. Chordates, because both are deuterostomes whose blastopore becomes the anus.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

In an embryo, the first opening (blastopore) develops into the mouth. To which group does the animal belong?

- A. Vertebrates only
- B. Cnidarians, such as jellyfish
- C. Deuterostomes, such as echinoderms and chordates
- D. Protostomes, such as annelids and arthropods

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Hikers on a trail maintained by a state park bring a teacher a small animal found on a deer. It has eight legs and two body regions, and it was attached near the deer's ear for about two days. How should it be classified?

- A. Crustacean
- B. Annelid, because it feeds on blood like a leech
- C. Arachnid
- D. Insect

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A tunicate (sea squirt) larva swims with a tail and has a notochord and dorsal nerve cord. The adult attaches to a dock piling, loses the tail and notochord, and filters water through slits. Divers counted 60 of them on one piling. How should tunicates be classified?

- A. As vertebrates, because they have a nerve cord
- B. As sponges, because adults filter water while attached
- C. As chordates, because they have all chordate traits at some life stage
- D. As non-chordates, because the adult lacks a notochord

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A nature center that serves about 12,000 visitors a year keeps a frog, a turtle, and a songbird. The frog's tank was upgraded last year. Which animals produce an amniotic egg that lets them reproduce away from water?

- A. Only the songbird
- B. The turtle and the songbird
- C. All three, since all three lay eggs
- D. Only the frog

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A vertebrate must return to water to lay eggs, which hatch into gill-breathing young. Which group is it most likely in?

- A. Birds
- B. Amphibians
- C. Mammals
- D. Reptiles

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A class raises insects in two containers. In one, the young look like small wingless adults and molt several times. In the other, the young are worm-like, then form a case before emerging as winged adults. The containers sit by a sunny window. Which description is correct?

- A.** The first shows incomplete metamorphosis; the second shows complete metamorphosis.
- B.** Both show complete metamorphosis because both molt.
- C.** The first shows complete metamorphosis; the second shows incomplete.
- D.** Both young are larvae.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A zoo keeps a 5-kilogram monitor lizard and a 5-kilogram fox in exhibits built in 2015. Keepers note the lizard basks under a heat lamp every morning. Which animal needs more food per week, and why?

- A.** About the same, because they weigh the same.
- B.** The lizard, because its body is always cold.
- C.** The fox, because endotherms burn energy to make body heat.
- D.** The lizard, because it must stay warm under the lamp.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

An animal's body temperature is 38°C when the air is 10°C and 39°C when the air is 30°C. What is it?

- A.** An endotherm
- B.** An ectotherm
- C.** An ectotherm, because it was warmer when the air was warmer
- D.** An amphibian

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

At a public aquarium ~~that opened in 1998~~, a volunteer shows visitors a jellyfish tank next to a flatworm display. ~~The jellyfish tank holds 900 liters of seawater.~~ Which statement correctly compares the two body plans?

Target word: compare symmetry **Decides:** Jellyfish are radial; flatworms are bilateral with a head end.

A	Correct	Cnidarians are radial; flatworms have one mirror plane and cephalization.
B	Inverted operation	Swaps the two body plans.
C	Absolute language	Habitat does not remove symmetry; most animals have one of the two types.
D	Adjacent category	Being an invertebrate says nothing about symmetry.

2. Answer B Concept 2

A teacher sets out three preserved specimens ~~bought from a supply company in the spring~~: a planarian (flatworm), a roundworm, and an earthworm. Which list correctly names each one's body cavity type?

Target word: match worms to cavity type **Decides:** Flatworm = none; roundworm = pseudocoelom; earthworm = true coelom.

A	Absolute language	"Worm" covers several unrelated phyla with different cavities.
B	Correct	The three worms span the three cavity types in that order.
C	Inverted operation	Reverses the order.
D	Confusion set	Roundworms have a pseudocoelom, not a fully lined coelom.

3. Answer D Concept 2 · transfer

An animal has a fluid-filled body cavity lined only partly by mesoderm. Which animal fits?

Target word: animal from cavity description **Decides:** Partly lined cavity = pseudocoelom = nematode.

A	Dropped element	Sponges lack true tissues, so they have no mesoderm-lined cavity of any kind.
B	Adjacent category	Flatworms have no body cavity at all.
C	Confusion set	Annelids have a fully lined coelom.
D	Correct	Nematodes are the classic pseudocoelomates.

4. Answer D Concept 3

A student notices that sea stars are invertebrates with a spiny skin, and asks which group they are most closely related to. ~~The class recently watched a documentary about coral reefs.~~ Based on embryo development, what is the best answer?

Target word: closest relatives of echinoderms **Decides:** Echinoderms and chordates are both deuterostomes.

A	Camouflage	Habitat from the documentary detail does not decide relatedness.
B	Inverted operation	Blastopore-to-mouth is the protostome pattern.
C	Adjacent category	Arthropods are protostomes; a hard covering is not a shared-ancestry trait here.
D	Correct	Shared deuterostome development links echinoderms and chordates.

5. Answer D Concept 3 · transfer

In an embryo, the first opening (blastopore) develops into the mouth. To which group does the animal belong?

Target word: group from blastopore fate **Decides:** Blastopore becomes mouth = protostome.

A	Confusion set	Vertebrates are deuterostomes.
B	Adjacent category	Cnidarians are diploblastic and outside the protostome/deuterostome split.
C	Inverted operation	In deuterostomes the blastopore becomes the anus.
D	Correct	“Proto-stome” means first mouth.

6. Answer C Concept 4

Hikers ~~on a trail maintained by a state park~~ bring a teacher a small animal found on a deer. It has eight legs and two body regions, and it was attached near the deer’s ear ~~for about two days~~. How should it be classified?

Target word: arthropod class **Decides:** Eight legs and two body regions.

A	Adjacent category	Crustaceans have two pairs of antennae and are mostly aquatic.
B	Camouflage	Feeding on the deer does not decide the phylum; jointed legs make it an arthropod.
C	Correct	Ticks, like spiders, have eight legs and two body regions.
D	Confusion set	Insects have six legs and three body regions.

7. Answer C Concept 5

A tunicate (sea squirt) larva swims with a tail and has a notochord and dorsal nerve cord. The adult attaches to a dock piling, loses the tail and notochord, and filters water through slits. ~~Divers counted 60 of them on one piling.~~ How should tunicates be classified?

Target word: classify an animal with larval-only traits **Decides:** Chordate traits need only appear at some stage of life.

A	Adjacent category	Vertebrates are a chordate subgroup with a backbone, which tunicates lack.
B	Confusion set	Filter feeding while stationary is shared by unrelated groups.
C	Correct	Chordate status rests on traits present at any stage.
D	Absolute language	Assumes the traits must last into adulthood.

8. Answer B Concept 6

A nature center ~~that serves about 12,000 visitors a year~~ keeps a frog, a turtle, and a songbird. ~~The frog’s tank was upgraded last year.~~ Which animals produce an amniotic egg that lets them reproduce away from water?

Target word: which animals are amniotes **Decides:** Reptiles and birds are amniotes; amphibians are not.

A	Dropped element	Leaves out reptiles, which are also amniotes.
B	Correct	Both are amniotes; their eggs carry water-holding membranes.
C	Absolute language	Frog eggs lack an amnion and dry out on land.
D	Inverted operation	The frog is the one that needs water to reproduce.

9. Answer B Concept 6 · transfer

A vertebrate must return to water to lay eggs, which hatch into gill-breathing young. Which group is it most likely in?

Target word: group from reproductive need **Decides:** Needing water for eggs = non-amniote tetrapod.

A	Inverted operation	Birds are amniotes that lay shelled eggs on land.
B	Correct	Amphibian eggs lack an amnion and larvae use gills.
C	Adjacent category	Mammals are amniotes; most do not lay eggs.
D	Confusion set	Reptiles lay amniotic eggs on land.

10. Answer A Concept 7

A class raises insects in two containers. In one, the young look like small wingless adults and molt several times. In the other, the young are worm-like, then form a case before emerging as winged adults. ~~The containers sit by a sunny window.~~ Which description is correct?

Target word: metamorphosis type in each container **Decides:** A pupal case means complete metamorphosis.

A	Correct	Nymphs resemble adults; a pupa marks complete metamorphosis.
B	Adjacent category	Molting occurs in both types; only the pupa separates them.
C	Inverted operation	Swaps the two patterns.
D	Confusion set	Young that look like small adults are nymphs, not larvae.

11. Answer C Concept 8

A zoo keeps a 5-kilogram monitor lizard and a 5-kilogram fox ~~in exhibits built in 2015~~. Keepers note the lizard basks under a heat lamp every morning. Which animal needs more food per week, and why?

Target word: food needs of endotherm vs. ectotherm **Decides:** The fox makes its own body heat.

A	Dropped element	Ignores the metabolic difference.
B	Absolute language	Ectotherms can be quite warm when the environment is warm.
C	Correct	Endotherms need several times the food of a same-size ectotherm.
D	Inverted operation	The lamp supplies the heat, so the lizard spends less energy.

12. Answer A Concept 8 · transfer

An animal's body temperature is 38°C when the air is 10°C and 39°C when the air is 30°C. What is it?

Target word: classify from temperature data **Decides:** Body temperature stays nearly constant as air changes.

A	Correct	It holds its temperature steady regardless of the air.
B	Inverted operation	An ectotherm's temperature would track the air.
C	Camouflage	A 1°C rise against a 20°C swing is steady regulation.
D	Adjacent category	Amphibians are ectotherms.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Radial vs. bilateral symmetry	Radial animals (cnidarians, adult echinoderms) can be cut into matching halves along many planes; bilateral animals have one plane, a head end, and usually move forward.	Classifying sea stars as related to jellyfish because both are radial as adults.	A body plan description and a question about the number of mirror planes or a head.
2	Body cavities	Acoelomates (flatworms) have no body cavity; pseudocoelomates (roundworms) have a cavity only partly lined by mesoderm; coelomates (annelids, mollusks, arthropods, chordates) have a fully lined coelom.	Assuming all worms share the same body cavity type.	“Flatworm,” “roundworm,” or “segmented worm” next to a question about the cavity.
3	Protostomes vs. deuterostomes	In protostomes the blastopore becomes the mouth; in deuterostomes it becomes the anus. Echinoderms and chordates are deuterostomes.	Grouping echinoderms with mollusks or arthropods because they are invertebrates.	“Blastopore” or a question about which phyla are most closely related to chordates.
4	Arthropod classes	Insects have three body regions and six legs; arachnids have two body regions and eight legs; crustaceans have two pairs of antennae.	Calling spiders and ticks insects.	A leg count or body-segment count in the stem.
5	Chordate traits	Chordates have a notochord, dorsal hollow nerve cord, pharyngeal slits, and post-anal tail at some stage of life, not necessarily as adults.	Thinking a chordate must keep all four traits or must have a backbone.	An animal that shows the traits only as a larva or embryo.
6	The amniotic egg	Reptiles, birds, and mammals are amniotes; their embryos develop inside membranes that hold water, which freed reproduction from water. Amphibians are not amniotes.	Assuming any egg-laying land vertebrate has an amniotic egg.	A question about which group can reproduce away from water.
7	Complete vs. incomplete metamorphosis	Complete metamorphosis goes egg-larva-pupa-adult; incomplete goes egg-nymph-adult, with nymphs resembling small adults.	Calling any young insect a larva.	A pupa stage, or young that look like wingless adults.
8	Ectotherms vs. endotherms	Endotherms generate body heat metabolically and need far more food; ectotherms rely on outside heat and their body temperature tracks the environment.	Equating ectotherm with “cold-blooded” and assuming its body is always cold.	Food needs, basking, or body temperature compared with air temperature.

Course scope (public description): Zoology provides graduate students seeking licensure or endorsement and/or their MA degree in secondary science education with an introduction to the field of zoology. Zoology includes the study of major animal phyla emphasizing characteristics, variations in anatomy, life cycles, adaptations, and relationships among the animal kingdom. A prerequisite for this course is Introduction to Biology.